

TOSHAL NARENDRA ZAMBARE

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PROFESSIONAL SUMMARY

Versatile B.E. student in Artificial Intelligence & Data Science with hands-on experience building production-grade AI/ML systems, full-stack SaaS platforms, cloud-native architectures, and IoT solutions. Strong foundation in Data Structures & Algorithms, microservices design, real-time systems, and containerized deployments. Passionate about bridging intelligent algorithms with beautiful, high-performing user interfaces. Cloud Lead at Google Developer Groups (GDG) on Campus at MET.

EDUCATION

Bachelor of Engineering (B.E.) in Artificial Intelligence & Data Science 2023 – 2027 (Expected)
Savitribai Phule Pune University (SPPU) — MET Institute of Engineering, Nashik **CGPA: 8.7 / 10.0**

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Computer Networks, Machine Learning, Deep Learning, Natural Language Processing, Cloud Computing, Operating Systems

Higher Secondary School (Class 12 — Science) 2023
Ord. Factory High School & Junior College, Ambarnath **Percentage: 72.67%**

Secondary School (Class 10) 2021
Ord. Factory High School & Junior College, Ambarnath **Percentage: 91.40%**

TECHNICAL SKILLS

Languages: C++, Python, C, JavaScript (ES6+), TypeScript, Java, Dart, C#, SQL, PHP, HTML5, CSS3

AI/ML & Data Science: PyTorch, TensorFlow, Keras, OpenCV, MediaPipe, LangChain, HuggingFace Transformers, Sentence-Transformers, NEAT (NeuroEvolution), Prompt Engineering, RAG Pipelines

Backend & APIs: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets (Socket.io), Server-Sent Events (SSE), Celery, Redis, JWT/OAuth 2.0 Authentication

Frontend & Cross-Platform: React.js, React Native, Redux, Expo, Three.js, Vite, WebGL

Databases & Storage: PostgreSQL, MongoDB, MySQL, Firebase/Firestore, Qdrant (Vector DB), MinIO (S3 API), Backblaze B2

DevOps & Cloud: Docker, Docker Compose, Nginx, AWS (Lambda, SQS, SNS, EC2, IAM, CloudWatch, Billing), Google Cloud Platform (Cloud Build), Vercel, Render, GitHub Actions, CI/CD

Game Development & IoT: Unity3D, Pygame, Arduino, ESP32, Relay Circuits, Blynk IoT

Core CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming, System Design, Microservices Architecture, DBMS, Computer Networks

WORK EXPERIENCE

Deep Learning & Full Stack Intern Dec 2025 – Feb 2026
AI Leela, Nashik

- Designed and developed **XeroxSaaS** — a production AI-powered SaaS platform for automated document processing using the MERN stack with real-time WebSocket notifications.
- Architected a containerized microservices backend using **FastAPI**, **PostgreSQL**, and **Redis**, integrating OAuth 2.0 for automated multi-provider data sync (Google Drive, OneDrive).
- Engineered a high-throughput asynchronous data ingestion pipeline using horizontally scaled **Celery workers** to extract, parse, and archive multi-format files into **MinIO** object storage.
- Implemented **Server-Sent Events (SSE)** and REST endpoints to stream low-latency analytics and AI chat responses to a **React/Redux** frontend.
- Built **LangChain-based AI agents** for intelligent document conversion, summarization, and page-level analysis using vector embeddings.
- Containerized all services with **Docker** and deployed on **Render**, **Vercel**, and **MongoDB Atlas**; currently active in local production use.

PROJECTS

— Full-Stack & SDE Projects —

XeroxSaaS — AI-Powered Document Processing SaaS 2026
Technologies: React.js, Node.js, Express.js, MongoDB Atlas, Socket.io, Backblaze B2, Docker

- Built a full-stack SaaS platform with multi-format document uploads, automated PDF normalization, and smart page counting with color detection.
- Implemented real-time order tracking, dynamic status notifications, and shop dashboard analytics using **Socket.io** WebSockets.

- Integrated cloud object storage (Backblaze B2) with automated file cleanup pipelines; deployed live at anti-print.vercel.app.

Scribble — Containerized Real-Time Multiplayer Drawing Game

2026

Technologies: Node.js, Express.js, Socket.io, HTML5 Canvas, Docker, Docker Compose

- Architected a low-latency, real-time multiplayer drawing game broadcasting canvas events to concurrent clients via WebSocket rooms.
- Developed room management, turn-based game state synchronization, word selection, scoring engine, and automated chat-based guess validation.
- Containerized client and server microservices using **Docker** and orchestrated multi-container deployment with **Docker Compose**.

Docview — Cross-Platform Mobile Document Viewer

2026

Technologies: React Native, Expo, JavaScript, Custom Android Native Module

- Developed a cross-platform mobile document viewer supporting file system navigation, search, and deep linking via custom intent handling.
- Built a custom Android Native Intent module (`withIntentDataModule`) to intercept system file-open requests and render documents with optimized memory handling.

Interactive 3D Portfolio Website

2026

Technologies: JavaScript, Three.js, Vite, Node.js, Express.js, Vercel

- Built an interactive portfolio website featuring WebGL-powered 3D scenes, animated particle backgrounds, and smooth scroll-driven transitions.
- Implemented a functional terminal emulator with custom commands, an admin panel with JWT authentication for content management, and Vercel serverless functions.
- Live at toshal.space.

Snake & Ladders — Multiplayer Board Game with Graph Algorithms

2025

Technologies: Python, Pygame, Graph Theory, BFS/DFS

- Developed a multiplayer Snake & Ladders game with a graphical UI using Pygame, featuring animated dice rolls and token movement.
- Implemented the game board as a graph data structure and used **BFS** to compute optimal move sequences and win probability analysis.

Online Quiz System — Full-Stack Web Application

2025

Technologies: PHP, MySQL, HTML/CSS, JavaScript, PHPMailer

- Built a full-stack quiz platform with role-based access (student/teacher), question bank management, timed quizzes, and automated grading.
- Implemented user authentication with password reset via email (PHPMailer), session management, and MySQL stored procedures.

Job Application Management System

2025

Technologies: PHP, MySQL, HTML/CSS, JavaScript

- Developed a database-driven job portal with CRUD operations for job postings and applicant tracking.
- Designed normalized ER schema, implemented complex SQL queries with joins, subqueries, and triggers for automated data validation.

— AI/ML & Data Science Projects —

Data Ingestion Platform — RAG-Powered Document Intelligence

2025

Technologies: FastAPI, React, Redux, PostgreSQL, Redis, Celery, MinIO, Qdrant, Docker, Nginx

- Built an enterprise-grade data ingestion system with OAuth 2.0 integration for Google Drive and OneDrive, enabling automated multi-provider file sync.
- Engineered an async ingestion pipeline with Celery workers processing multi-format files (PDF, DOCX, XLSX, images) with OCR extraction.
- Implemented RAG (Retrieval-Augmented Generation) chat interface backed by **Qdrant** vector database and sentence-transformer embeddings.
- Orchestrated 7+ Docker containers (backend, frontend, workers, Redis, PostgreSQL, MinIO, Nginx) via Docker Compose.

Alert Fatigue Triage Engine — AI-Powered Security Alert System

2026

Technologies: Python, FastAPI, React.js, PostgreSQL, Docker, MCP Server, LLM Agent

- Built an intelligent alert triage system that uses LLM-powered agents to classify, prioritize, and auto-remediate security alerts, reducing alert fatigue.
- Developed a React.js dashboard for real-time alert visualization and a FastAPI backend with PostgreSQL persistence.
- Implemented an MCP (Model Context Protocol) server to expose alert data to AI agents for autonomous decision-making.
- 1st Place Winner — GDG Nashik Cricketthon Hackathon, 2026.

NEAT-Based Flappy Bird — Neuroevolution AI Game

2026

Technologies: Python, NEAT-Python, Pygame, Plotly

- Implemented a Flappy Bird clone with AI agents trained using **NEAT** (NeuroEvolution of Augmenting Topologies) algorithm.
- Built a Plotly-based analytics dashboard visualizing genome fitness evolution, species diversity, and neural network topology across generations.
- Achieved perfect gameplay (infinite score) within 5–10 generations using feedforward neural networks with 3 inputs and 1 output.

Handsign Language Recognition — Real-Time Computer Vision

2025

Technologies: Python, OpenCV, MediaPipe, TensorFlow, scikit-learn

- Built a real-time hand sign language detection system recognizing ASL letters (A, B, C) using webcam feed.
- Used **MediaPipe Hands** for 21-point landmark extraction and trained a classification model on custom gesture datasets.
- Achieved real-time inference at 30+ FPS with bounding box visualization and confidence scoring.

Handwritten Digit Recognition — Neural Network Classifier

2025

Technologies: Python, TensorFlow, Keras, NumPy, Jupyter Notebook

- Built and trained a deep neural network for MNIST handwritten digit classification achieving 98%+ accuracy.
- Implemented data preprocessing, model architecture design, training loops, and evaluation with confusion matrices.

AI Voice/Text Communication Game — Unity NPC System

2025

Technologies: Unity3D, C#, Python, Flask, WebSocket, LLM API

- Developed a Unity game where players communicate with AI-powered NPCs via voice and text in natural language.
- Built a Python Flask server handling speech-to-text, LLM-based dialogue generation, and text-to-speech synthesis for immersive NPC conversations.
- Implemented WebSocket communication between Unity client and Python AI server for low-latency real-time interactions.

— Cloud & DevOps Projects —

AWS Student Kill Switch — Serverless Cost Protection

2026

Technologies: Python, AWS Lambda, SQS, SNS, IAM, CloudWatch, Boto3

- Built a serverless AWS cost-protection system using **Lambda + SQS + SNS** event-driven architecture to automatically detect and terminate runaway resources.
- Implemented IAM least-privilege policies, CloudWatch billing alarms, and automated EC2/RDS instance shutdown workflows.
- Included a local Python worker fallback for environments without Lambda access; complete CLI setup and teardown guides.

Google Cloud CI/CD Pipeline

2025

Technologies: Python, Google Cloud Build, Cloud Run, Docker, GitHub

- Configured continuous integration and deployment pipelines using **Google Cloud Build** with automated Docker image builds and Cloud Run deployments triggered by GitHub push events.

— IoT & Embedded Systems Projects —

IoT Home Automation System

2025

Technologies: Arduino, C++, ESP32/NodeMCU, Relay Modules, Blynk IoT

- Designed a smart home automation system using microcontrollers (Arduino/ESP32) controlling lights and fans via relay circuits.
- Implemented remote control via the Blynk IoT mobile app with Wi-Fi connectivity for real-time appliance switching.

IoT Environmental Monitoring Web Dashboard

2025

Technologies: Python, Flask, HTML/CSS/JS, ESP32, Sensors

- Built a web-based dashboard for real-time environmental monitoring (temperature, humidity, air quality) with data streamed from ESP32 sensors.

— DSA & Algorithmic Projects —

DSA Problem Solutions Collection

2025 – Present

Languages: Python, C++

- Maintained a curated collection of optimized solutions for Data Structures & Algorithms problems covering arrays, linked lists, trees, graphs, dynamic programming, greedy algorithms, and backtracking.

LEADERSHIP & ACHIEVEMENTS

- **1st Place** — Cricketthon Hackathon, Google Developer Groups (GDG) Nashik

Aug 2026

- **3rd Place** — HackFusion Hackathon 2025
- **Cloud Lead** — Google Developer Groups (GDG) on Campus at MET, Nashik 2024 – 2025
- **Flutter Co-Lead** — GDG on Campus, Flutter Development Wing at MET 2024 – 2025
- **Open-Source Contributor** — Hacktoberfest 2024

CERTIFICATIONS

- **Python for Data Science & DSA** — NPTEL 2024
- **Data Structures & Algorithms in Python** — NPTEL 2024
- **AWS Cloud Foundations** — AWS Cloud Internship 2024
- **Java Programming Certification** — Oracle Academy 2024
- **AI-ML Virtual Internship** — EduSkills × Google for Developers 2024
- **Name Generator with TensorFlow** — Coursera 2024
- **C++, C, Python Programming** — Spoken Tutorial, IIT Bombay 2024 – 2025

INTERESTS & ACTIVITIES

Open-Source Development, Competitive Programming, Hackathons, Technical Blogging, Community Leadership, Game Development, IoT Prototyping, Cloud Architecture